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Crystal Growth from Solutions

BY W. F. BERG, PH.D. (BERLIN), PH.D. (MANCHESTER)

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[Plate 2]

INTRODUCTION

The purpose of the present paper is to provide an experimental contribution towards the knowledge of the mechanism of crystal growth from solutions.

Certain details about the mechanism of crystal growth have been studied in the case of one-component systems, i.e. crystals growing from the melt or from the vapour. Very little is known about crystal growth from solutions. Yet, although the one-component system is the simpler one, special interest is attached to the two-component system, because of its technical applications. In chemical industry crystals are grown from solutions on a big scale. A closer knowledge of the mechanism of crystal growth would enable one to apply the conditions, e.g. temperature or degree of supersaturation, most favourable for the process in question. This is, as yet, done purely empirically.

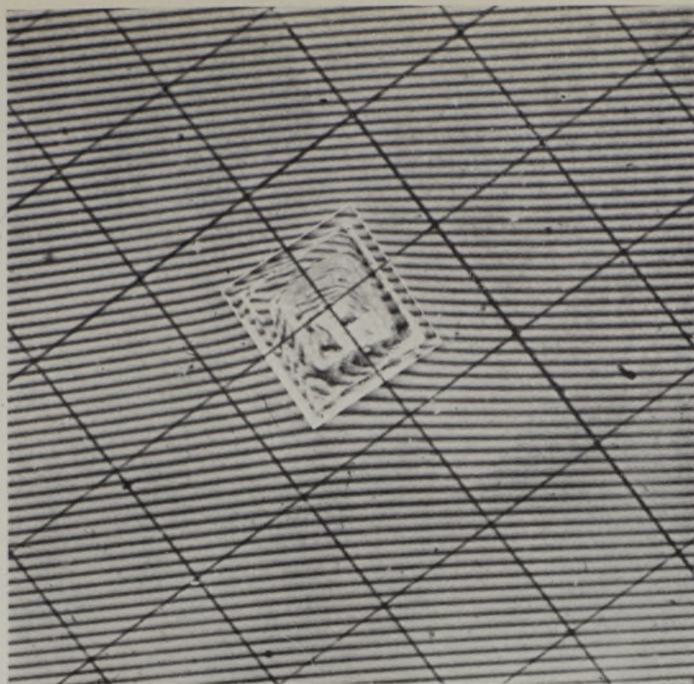
The problem covers a wide field, and an attempt to answer what was thought to be one of the first questions to be asked will be described below.

This is the question of the connexion between the rate of growth and the concentration of the solution in contact with the growing crystal face. It will be seen that this question cannot be answered yet, mainly because impurities play too big a part in determining the rate of growth. But the method applied seems to reveal certain facts about the mechanism of crystal growth from solutions.

The concentration in actual contact with a dissolving or growing crystal has, since Nernst (1904; see also Brummer 1904), always been thought of as being the saturation concentration. The rate of growth was then supposed to be proportional to the degree of supersaturation. Several investigators have tried to prove these ideas experimentally. Their method generally consisted of putting a known amount of seed crystals into a solution of known supersaturation and of separating the crystals from the solution after a certain time and of analysing it. The solution was kept in rapid motion, and it was hoped to get solution of the measured concentration right up to the crystal faces, so that the concentration gradient became very big and diffusion processes could be neglected. It seems very difficult to judge whether this has been achieved or not. It is still possible that in these experiments the rate of growth was mainly governed by the rate of diffusion through a thin stagnant film of solution adhering to the crystal surface. The results of the different papers are not in agreement. The rate of dissolution was found by Noyes and Whitney (1897) to be proportional to the degree of unsaturation. Marc (1905, 1908*a*) found the rate of growth to be proportional to the square of the supersaturation. His results were not confirmed completely by later investigators (Le Blanc and Schmandt 1911, 1913; see also Jenkins (1925) and Roller (1932)). Experiments on the influence of dyes on crystal growth led Marc (1908*b*, 1910, 1911*a*) to assume the existence of an adsorbed layer on the surface of a crystal in contact with its solution. This layer forms an intermediate state between the liquid and the solid. Molecules go into this layer and stay there until the conditions become favourable for them to freeze into the lattice. Dye molecules are adsorbed by the crystal, thereby changing the rate of growth of the particular faces adsorbing them and thus changing the habit of the crystal.

Experiments by Volmer (1932) performed, however, on crystals growing from the vapour or from the melt, demonstrated the existence of such an adsorbed layer and its importance for the mechanism of crystal growth. It was shown that the molecules in the adsorbed layer have a much higher mobility than in the melt.

It appears that many questions concerning crystal growth need clearing *a*



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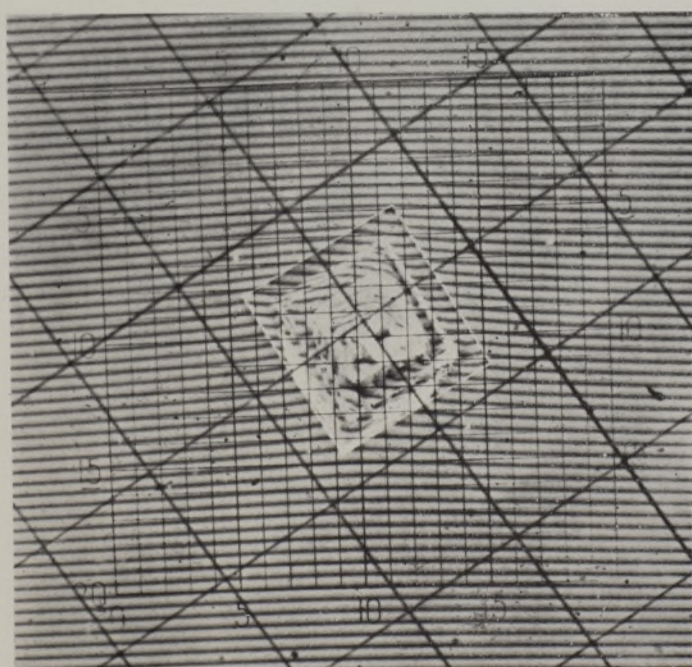

FIG. 2—Shows two out of a series of photographs from a growing crystal. The big graticule was photographed at the same time as the crystals, the length of the sides of the squares corresponding to $\frac{1}{2}$ mm. The small graticule was attached afterwards for measuring purposes. *a* is the photograph from which the diagram fig. 6 was obtained. The photographs are fixed in the same position as the diagram: the lines are running horizontally. The photographs are negatives.

(*Facing p.* 80)

up, especially in view of the existing discrepancies. Complete changes in the method of attack and experiments on single crystals seem to be called for, since diffusion effects have to be eliminated or taken into account.

Method

A method in which the concentration can be measured everywhere in the neighbourhood of a growing or dissolving crystal has been suggested to the writer by Mr T. R. Scott, of I.C.I. (Alkali), Ltd., Winnington. In this the problem is first reduced to a two-dimensional one, by placing a thin prismatic crystal of rectangular cross-section between two horizontal glass plates, which stop growth in the third dimension. Secondly, convection currents are eliminated by doing the experiments on a microscopic scale. Thus the transport of molecules is purely a two-dimensional diffusion process. Thirdly, the two glass plates, forming a slight wedge, are half-silvered and illuminated with semi-parallel monochromatic light, thus producing a system of interference fringes. These fringes are straight if the wedge is optically perfect and if the refractive index of the matter in the wedge is constant. They are curved if the refractive index changes from place to place and thus provide a simple means of measuring the refractive index. The refractive index will vary if a growing crystal takes molecules from the surrounding liquid. Since top and bottom of the crystal touch the glass plates, symmetry considerations show that the diffusion process will be identical in layers parallel to the plates. Thus the concentration distribution can be measured right up to the crystal face, if the refractive index is known in terms of the concentration. Measurements by Miers and Isaac (1906) give the necessary figures for sodium chlorate: the refractive index varies linearly with the concentration. Sodium chlorate was chosen for the following experiments, because it crystallizes in rectangular prisms, and is a cubic crystal. Supersaturated solutions of it are also easily obtained.

Apparatus

The apparatus consisted essentially of a photomicrographic outfit. There are, however, details which seem worth mentioning (fig. 1).

A "Patna" microscope by Watson was used, which had a specially made short and wide body tube, to avoid cutting down the field of view. Microscope, camera, and source of light were all mounted on the cylindrical bed of a 4 in. Drummond lathe. This lathe proved extremely convenient for optical purposes. The lens used for photographing was a *F* 3·5 Ross Xpress of 5 cm. focal length. No eyepiece was used for photographing.

It appeared desirable to photograph a system of fiducial marks at the same time as the growing crystal and its surrounding fringe system. This would facilitate comparison of different plates and would also give at once the magnification. For this purpose a luminous graticule was used, which was obtained by contact printing from an eyepiece micrometer graticule.

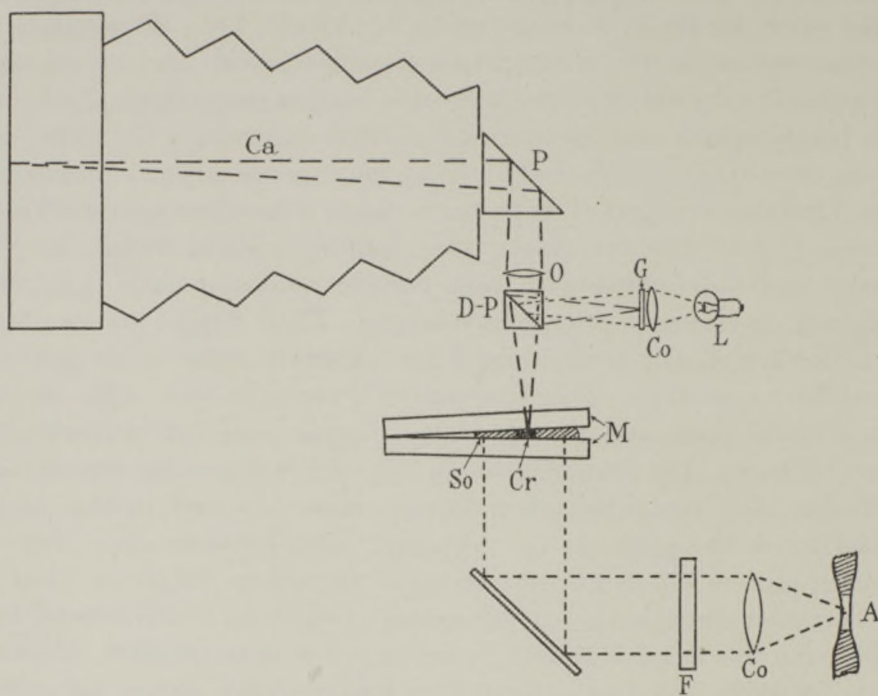

FIG. 1—Shows the apparatus for photographing the growing crystals and the surrounding interference system. *A* is the mercury arc, *Co* a condenser, *F* a filter, *M* are the two half-silvered mirrors with the crystal *Cr* and the solution *So* between them. *L* is a pea lamp, *Co* another condenser, *G* the graticule with light lines on a dark ground. *D–P* is the double prism acting as transparent mirror, *O* the object glass, *P* a right-angle prism and *Ca* the camera.

A slightly silvered mirror was fixed underneath the objective so that the object and the graticule could be seen at the same time. The mirror consisted of two right-angle prisms, one of which was slightly silvered, stuck together with canada balsam. The graticule was fixed in front of a small condenser and a pea lamp, and the whole system could be moved towards the mirror to focus it at the same time as the object.

The illumination for the interference phenomenon was provided by a small mercury arc taking 0·25–0·4 amp. The light was made monochromatic by the use of filters. The actual arc burned in a capillary which was so narrow that a symmetrical achromatic condenser made a beam of light sufficiently parallel to produce sharp interference fringes.

Photographs were taken on Ilford Hypersensitive Panchromatic plates, because of their high speed for the mercury green line $\lambda = 5461$ A. For visual observations when adjusting the apparatus this line was the brightest and most convenient one. Exposure times ranged from 2 to 30 sec., applying a magnification of about 10 linear.

The good quality of the interferometer plates and their silvering was essential to success. The plates were 1 by 3 in. and $\frac{1}{4}$ in. thick, worked by Bellingham and Stanley "correct to $\frac{1}{4}$ wave-length". The silvering was done by cathode sputtering (Tolansky and Lee 1936).

Experiments

The actual experiment consisted in placing one seed crystal in a drop of supersaturated solution between the two optical flats and taking a series of photographs of the crystal and the interference system at known times.

Flat rectangular seed crystals about 1 mm. square and 0·1 mm. thick were obtained by slowly evaporating a drop of sodium chlorate solution on a piece of glass plate. The degree of supersaturation of the solution was obtained by weighing. The crystalline material was supplied by the British Drug Houses as having the following analysis:

Chloride	0·02 %
Sulphate	Nil
Calcium	Nil
As_2O_3	0·8 part per million
Heavy metals, etc. ...	Extremely faint trace.

The main difficulty in the experiments was to avoid the formation of unwanted nuclei. To stop this several precautions were applied. The material was recrystallized twice, always discarding the first and the last portion. It was hoped that physical impurities would thus be removed. All the water used was distilled water and was filtered through filter crucibles before being used. Ordinary filter paper caused trouble by giving off fluff.

Despite all these precautions it was extremely difficult to conduct an experiment so that the seed crystal was the only one present. After many trials it was found that these unwanted crystals came from the surface of the seed crystal. It appears that when the solution round the growing crystal dries up on preparation of the seed crystal, the material of the last drop does not freeze into the structure of the crystal, but forms many very small crystals on top. One might perhaps get over this by making evaporation extremely slow. Another procedure was applied here. It consisted in washing the seed crystal just before it was used in a drop of very slightly unsaturated solution. The wet crystal was then transferred to one of the mirrors, the unsaturated solution soaked away with a piece of filter paper, the supersaturated solution applied with a small pipette and the whole covered with the second mirror. The utmost speed was necessary, otherwise more nuclei were formed.

The obvious disadvantage of getting rid of unwanted nuclei in this way is that the degree of supersaturation of the liquid surrounding the crystal at the start of the experiment is unknown, since it is not the same as the supersaturation of the solution one applies. Some of the unsaturated solution may still stick to the crystal and mix with the supersaturated liquid. It will be seen below how this difficulty was overcome by measuring the refractive index far away from the crystal.

Although few precautions were taken the room temperature during the experiments only changed a few tenths of a degree centigrade. The room temperature was measured each time a photograph was taken.

When the seed crystal with its surrounding supersaturated solution had been placed between the two mirrors, the whole arrangement was transferred to the microscope stage and observed, using monochromatic light. The illumination was adjusted till the fringes were sufficiently sharp, and when it was ascertained that no unwanted crystals were present the eyepiece used for visual observations was taken out and fringe system and graticule focused on the screen of the camera. The first photograph was usually taken 3–4 min. after the experiment was begun. This and the last photograph always included the boundary of the drop. Fringes appeared in the solution and in the neighbouring air, and by counting them over a distance the ratio of the refractive index of air and of the solution could be obtained. This is a simple means of measuring the absolute value of the concentration far away from the crystal, which had to be done, because owing to the method used the degree of supersaturation was not known at the start and would most certainly change during the experiment. The other photographs, six or eight in the course of 1 or 2 hr., were taken at different measured intervals with the crystal in the centre of the photographs.

The thickness of the wedge was found by measuring the thickness of the crystal after the experiment. This was done by removing the top mirror, drying the crystal superficially and focusing on the top and bottom of the crystal, using high magnification and the fine adjustment of the microscope. By comparing the photographs taken at different times it could be ascertained whether the thickness of the wedge had changed, i.e. whether the crystal had grown in the third dimension, and if so by how much. The fringes will move twice the number of fringe distances as the thickness changes in wave-lengths, thus providing a rather sensitive test. No appreciable change in thickness of the wedge was ever found.

Evaluation of the photographs and sources of error

Fig. 2 (plate 2) shows typical examples of the photographs obtained. The curvature of the fringes as they approach the crystal is clearly visible.

The shift of the fringes is now to be evaluated in terms of changes in refractive index and concentration. The optical path difference between a ray going straight through the wedge and one reflected once at the upper and once at the lower mirror is $2\mu_0 d$ and this has to equal $N\lambda$ for a bright fringe

$$2\mu_0 d = N\lambda. \tag{1}$$

If the fringe system is displaced by n fringe distances because μ_0 has changed to μ_1, we have

$$2\mu_1 d = (N+n)\,\lambda \tag{2}$$

and thus

$$\mu_0 - \mu_1 = -n\lambda/2d, \tag{3}$$

where μ_0 = the refractive of the supersaturated solution at a distance from the crystal,
μ_1 = the changed refractive index,
d = the thickness of the wedge,
N = an integer,
λ = the wave-length of the light.

λ is known and d is obtained experimentally. Thus, to obtain the concentration at any point on a fringe, it is only necessary to determine the displacement of the fringe at that point from the straight line formed when no crystal is present. If only relative concentrations are wanted, as, for example, if only the concentration gradient distribution is interesting, it is sufficient to measure and discuss the fringe displacement only without knowing the absolute value of the refractive index and of the concentration.

The concentration gradient can also be obtained more directly than by measuring the concentration distribution. One has only to measure the angle between the tangent to a deviated fringe at the required point and the original straight line, and the distance of the fringe from its neighbours. Since this method will be applied most frequently close to a crystal boundary it is more convenient to measure the spacing of the fringes along the crystal surface and the angles between the crystal and the fringes (see fig. 3). Let

x_N = x-coordinate of the Nth fringe,
a = a constant,
b = spacing of the fringes in the x-direction (undeviated),
γ = tan of the angle α between the z-axis and the undeviated fringes.

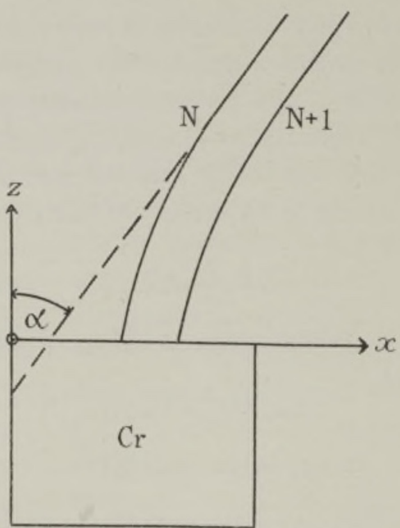

Fig. 3—Illustrates the calculation of the normal concentration gradient from the curvature and the distance of the fringes. The crystal *Cr* and two fringes N and $N+1$ are drawn, whose distance (undeviated) measured parallel to the x-axis is b. If the fringes were not deviated they would make the angle α with the z-axis.

We can then derive a formula for the component of the concentration normal to the crystal face $\partial c/\partial z$. This component is of greatest interest, since it will be shown below that it is proportional to the current of molecules going into the crystal. We find

$$\frac{\partial c}{\partial z} = \frac{\gamma}{\alpha} - \frac{1}{a}\left(\frac{dx}{dz}\right)_{z=0} \frac{b}{(x_N - x_{N-1})_{z=0}}. \tag{4}$$

Thus we have to measure the inclination of the fringes with respect to the crystal face $\left(\frac{dx}{dz}\right)_{z=0}$ and the fringe distance $x_N - x_{N-1}$, and the distance of the undeviated fringes b which can be obtained far away from the crystal, where the concentration is still the original one. The distribution of $\partial c/\partial z$ is then known apart from an unknown constant a. This method will not be used for quantitative results, since the accurate measurement of the angle is rather difficult, but it enables one to judge at a glance how the concentration gradients are distributed.

The measuring up of the diagrams was done in two ways. At first the diagram was brought into contact with a graticule of twenty-one horizontal "lines" and twenty-one vertical "columns", the lines being made to run parallel to the undeviated fringes far away from the crystal (see fig. 2). The positions of the fringes with respect to this graticule were measured by means of a microscope and an eyepiece micrometer scale. Later on more accurate measurements were done by means of a two-dimensional comparator. The eyepiece of the microscope of the comparator contained two parallel lines of variable distance, and in measuring the fringe was "caught" between those two lines. This procedure increased the accuracy appreciably, but produced no qualitative changes from the earlier method in the results given below.

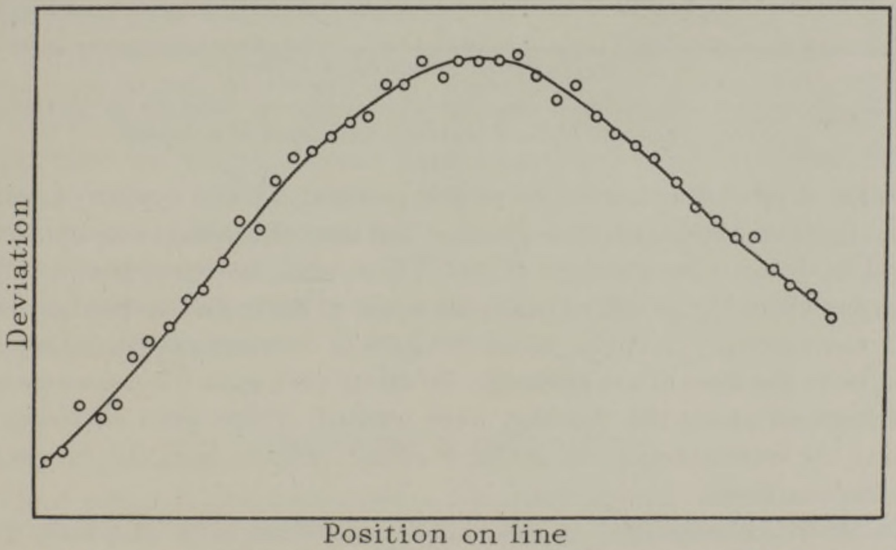

Fig. 4—Gives an example of the concentration distribution along a "line". Each point represents the position of a fringe against which is plotted the deviation of this fringe from straight-line configuration.

The deviations of the fringes, i.e. the difference between their position and the straight-line configuration as determined far away from the crystal, where the fringes were not yet shifted, was plotted against the position of the fringes along the lines. Fig. 4 shows a typical example. The positions of the fringes were also measured along the crystal surface (fig. 5). Obviously the latter values are of special interest.

Relative concentration diagrams were obtained from these curves in the following way. Those coordinates on a certain line were found, where the deviation had certain arbitrarily fixed values, the difference between these values being constant. Thus, for example, the coordinates may be found, where the deviation is 1, 2, 3, etc., fringe distances. The coordinates obtained in that way were plotted on a graph, which was a reproduction of the graticule with its twenty-one lines and twenty-one columns, and

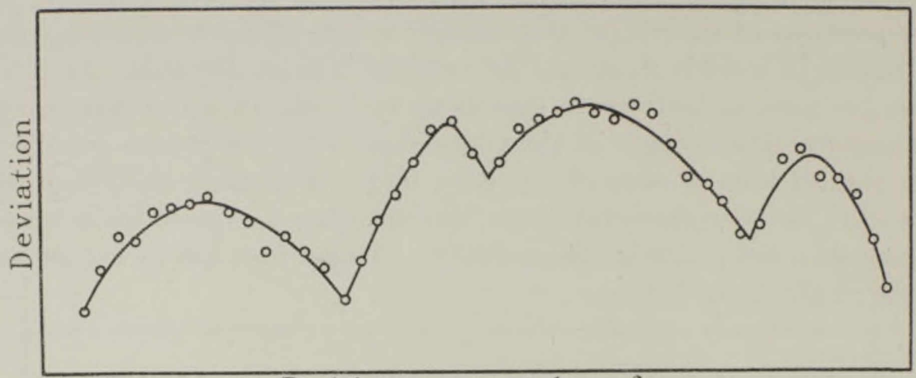

FIG. 5—Shows the concentration distribution in arbitrary units (fringe deviations of the solution in contact with the four faces of a crystal.

with the crystal drawn in in its proper position. Thus a system of points of constant concentration was obtained and lines of constant concentration could be drawn through these points. There were, however, places in the diagram where the points were too far apart to draw curves through them with any certainty, namely, where the lines of constant concentration run parallel to the lines of the graticule. To fill up such gaps the concentration distributions along the columns were wanted. These were obtained by taking the concentration values for a certain column from the curves for the various lines.

In the final concentration diagram the different sorts of points were marked in a different way, because they carry a different weight. The points on columns (●) are obtained by a twice-repeated smoothing out process (drawing a curve through points) and therefore carry the biggest weight, assuming that there are no systematic errors. Points on lines are subject to only one such process (○). Points on the crystal surface (·) carry even less weight, because each of them was measured separately and is therefore subject to the error attached to a single measurement.

The accuracy of these concentration diagrams is not as high as is desirable. This is mainly due to the fact that the flats used were only accurate to $\frac{1}{4}$ of a wave-length. Much trouble was caused by this fact, particularly in one case, where an inexplicable distortion of the fringe system was finally traced back to the existence of a pit in one of the flats. As soon as the crystal had grown across the pit and the fringes were in a different part of the wedge the distortion disappeared. It would have been quite impossible to obtain any results at all if the accuracy of the flats over the small areas used had not been actually much higher. The necessity of using the highest grade flats obtainable for any future experiments is emphasized by the fact that the accuracy of measuring the position of a fringe may under favourable conditions be as high as $\frac{1}{50}$ of a fringe distance, i.e. changes in the wedge thickness of $\frac{1}{100}$ of a wave-length would be detectable.

Systematic errors may arise from the heat of crystallization. The boundary of a growing crystal is a source of heat. Any local change in temperature would distort our fringe system. It can be shown by simple calculations, however, that in our case the rise in temperature on the crystal boundary will be between 10^{-4} and 10^{-2}° C. only.

Other errors may arise from the heat from the illuminant being absorbed by the crystal and its surroundings, resulting in a distortion of the fringes. As a check experiment a crystal was put into saturated solution, and it was found that the fringes showed hardly any observable curvature. What little shift there was was explained by evaporation from the boundaries of the drop, and the shift was the same all over the crystal, a point which will be important later on.

It seems justified to treat our concentration diagrams as if there were no systematic errors.

Discussion of the results

Fig. 6 shows a typical example of the concentration diagrams obtained.

We are now in a position to discuss our original question of how the rate of growth depends on the degree of supersaturation in contact with the crystal surface. It becomes clear at once by looking at the diagram fig. 6 and at fig. 5 that the question put forward in this simple way has no meaning, because the concentration in contact with the crystal varies from place to place. This has been found to be always the case and is of great importance in view of the older suggestion that a layer of definite concentration surrounds the whole crystal (Nernst 1904). The concentration was always found to be greater near the corners of the crystal than at the centres of its faces. Thus it seems a more appropriate question to ask how the rate of growth depends on the maximum, or minimum, or average, concentration in contact with the crystal face. But even such a question is, for experimental reasons, extremely difficult to answer. An example may illustrate this.

It has occasionally been found that two opposite, i.e. in our case

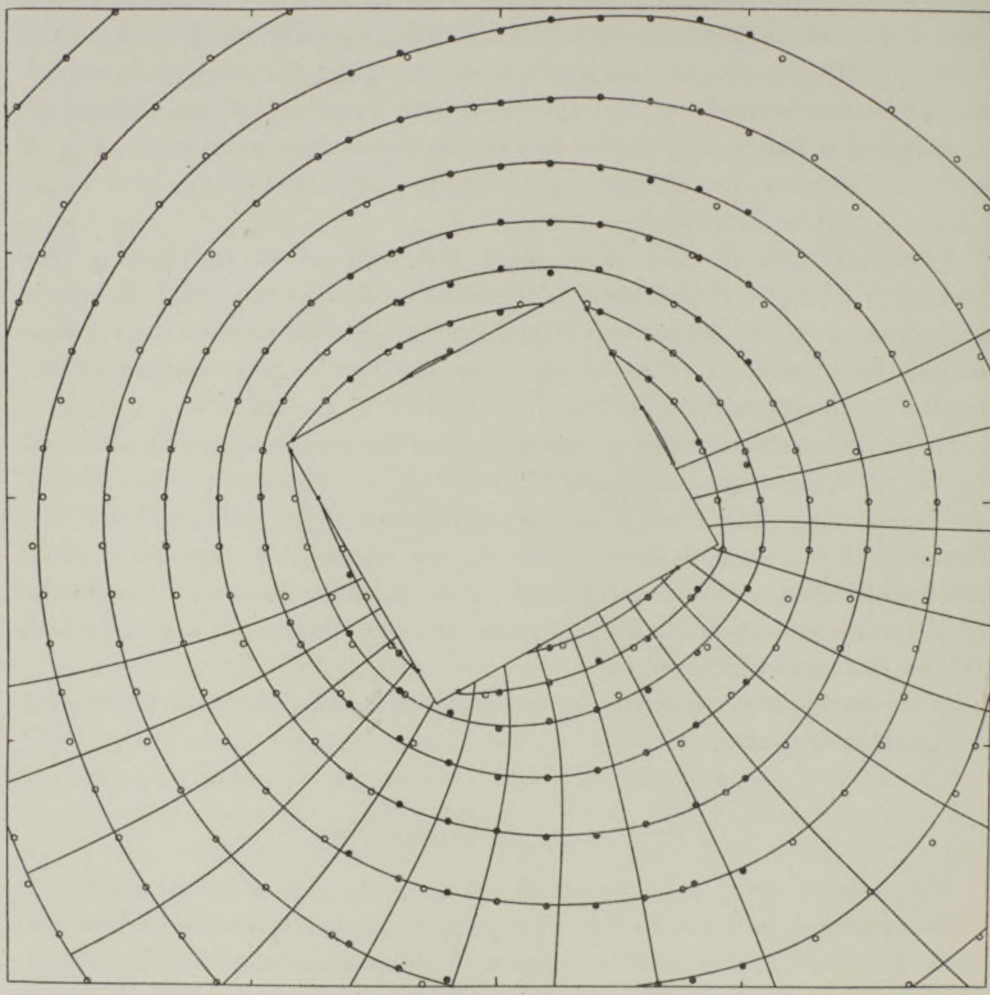

FIG. 6—Shows a typical concentration distribution round a growing crystal. The circles are points on "lines" (○), the large dots points on "columns" (●), the small dots (·) points measured on the crystal surface. "Lines" run horizontally, "columns" vertically in the diagram. Lines of flow are drawn in the lower part of the diagram. Note that the spacing of the lines of flow and of the lines of constant concentration is practically the same near the corners as at the centres of the faces. The photograph from which this diagram was obtained is shown in fig. 2*a*.

geometrically and crystallographically equal, faces of the same crystal, which were in contact with solution of the same degree of supersaturation at the start, grew at different rates. For this very reason the slower growing face was after a time in contact with more highly supersaturated solution than the other one. Small crystallization velocity means higher concentration on the crystal surface under otherwise similar conditions. The explanation is that the crystallization velocity is influenced by minute traces of impurities. Hence consistent results have not yet been obtained and it will not be easy to obtain them.

More interesting results were obtained when the diffusion gradients at different parts of the same crystal face were compared. In this connexion it was interesting to note that in our experiments stationary conditions were reached almost at once. Only during the first moments after the experiment had been started was there any appreciable movement of the fringes. The graticule photographed on each plate proved very useful here. Two successive photographs could be laid on top of each other, the graticules registering, and the shift of the fringes observed. If such a shift had occurred at all it was slight compared with the deviation of the fringes from the straight-line configuration. Exceptions were cases where the boundary of the drop of supersaturated solution was quite unsymmetrical with respect to the seed crystal. Then the solution on the one side lost its supersaturation much quicker than on the other side, and the fringes moved accordingly. It is self-evident that there will be a slight continuous shifting of the fringes all the time, since the dimensions of the drop are finite and the supply of molecules is therefore limited. The fact that stationary conditions prevail makes it quite simple to work out from our concentration diagrams the currents of molecules diffusing towards the crystal. A two-dimensional diffusion equation

$$\frac{\partial c}{\partial t} = -D\nabla^2 c \tag{5}$$

holds for the general case, but this is reduced to

$$\nabla^2 c = 0 \tag{6}$$

for our stationary conditions. The flow of molecules j is given by the equation

$$j = -D \operatorname{grad} c. \tag{7}$$

The diffusion coefficient for sodium chlorate has not yet been determined, but it can in principle be found from any set of concentration diagrams.

For the results described below the knowledge of the diffusion coefficient is not necessary.

The fact that stationary conditions can be assumed provides us with an easy means of checking the validity of the diffusion equation in our case, ensuring that no systematic errors fake our results. According to equation (6) the lines of flow and the lines of equal concentration are perpendicular to each other, and the spacing of the lines of flow is proportional to the spacing of the lines of equal concentration. The density of the lines of flow gives the flow per unit area.

These relations can, of course, only be verified graphically, since the equation cannot be solved analytically except for a few special boundary conditions. To prove the relations lines of flow were constructed as indicated above. They were found to run perpendicular to the lines of equal concentration fairly satisfactorily, if their spacing was made proportional to the spacing of the lines of equal concentration. The lines of flow in fig. 6 were obtained in this way. The accuracy with which this law was fulfilled gave little reason to doubt the validity of the diffusion equation for our case or of the concentration measured in the way described. Any deviations from the law were not of a systematic nature. Sometimes the lines of flow were a little too close, sometimes a little too far apart, which was explained by imperfections of the wedge. It seems thus justified to work out the flow of molecules from the concentration diagrams by applying the diffusion equation. But this result means more. It should be possible to construct the whole concentration diagram from the knowledge of the position of two neighbouring lines of constant concentration. Although one would never push it thus far this conclusion is certainly a great help in evaluating concentration diagrams, if due to local imperfections the diffusion equation does not seem to hold over the whole diagram.

We can now proceed to work out the current distribution over the crystal faces. A crystal face has generally the same rate of growth all over, as is evident from the fact that it remains plane, and only such faces were considered. This supposition is easily verified by inspecting the different photographs taken. It will be interesting to see whether the current of molecules flowing towards a crystal face is the same all over or not. If the flow of molecules per unit area of the crystal is the same at every point one would say that the molecules are taken out of the solution by the growing crystal just where they are wanted to produce plane crystal faces. If the flow is found to differ from place to place there must be another mechanism to make an even speed of growing possible.

The flow per square centimetre or, in our two-dimensional case, per unit length of the crystal is the quantity to be compared; this is proportional to the component of the concentration gradient normal to the crystal face. The concentration gradient is in turn inversely proportional to the spacing of the lines of constant concentration.

The result of such a comparison is, as is easily seen from the concentration diagram reproduced, that the component of the concentration gradient normal to the face does differ from place to place. We see that this concentration gradient is smallest near the corners and rises to a maximum at the centre of a face. The absolute value of the gradient itself does not differ much from place to place, at least not over one face, so that one is tempted to assume as boundary condition that the absolute gradient is constant. Higher accuracy will be necessary to confirm this. Since the lines of constant concentration cut the crystal at about 45° at the corners, the normal component of the gradient, i.e. the flow of molecules per unit area of the crystal surface, is deficient at the corners.

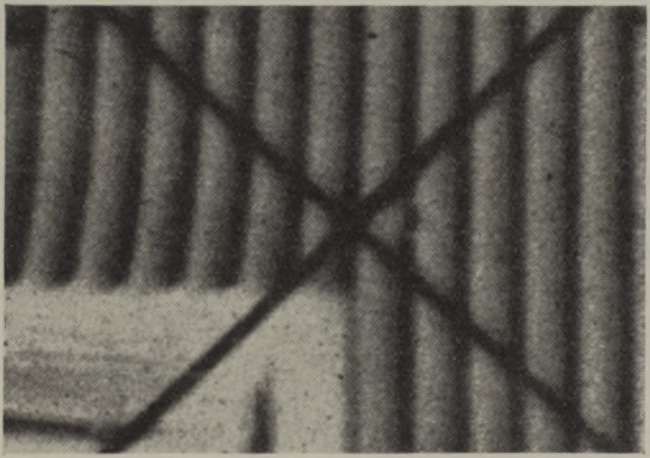

FIG. 7—Shows a much enlarged portion of a photograph of a crystal and its surrounding fringes to demonstrate the difference in angle between the fringes and the crystal surface near the corner and at the middle of the face.

The suspicion may arise that a systematic error in the evaluation has caused these results. But, apart from our discussion on possible errors, as has been shown above, there is the possibility of estimating the normal component of the concentration gradient from the curvature and the distances of the fringes, and if the results are true we should notice that the curvature and the spacing of the fringes is smaller near the corners of the crystal than at the centres of its faces. This is indeed the case as is demonstrated in fig. 7.

A new mechanism has to be assumed which enables the crystal to produce plane faces, although the current of molecules arriving at the corners is smaller than at the centres of the faces. The only possibility seems to assume a lateral surface flow of molecules from the centre of a face towards the corners. This current must flow in a very thin layer, because otherwise it would cause optical effects and would show up in our photographs by a shift of the fringes. This result has been obtained on all crystals so far measured. The decrease in the flow of molecules near the corners was found to be of the order of 25–50 % of the flow near the centres. That means that this percentage of the molecules going into the corner of a crystal must flow part of their way in the thin layer.

We are thus led to assume the existence on crystals in contact with their solutions of a layer which constitutes a transition between the solid and the liquid state. We may call it an adsorbed layer in order to signify that it is very thin without implying that it is a monomolecular layer as seems to be quite often the case with adsorbed layers. This adsorbed layer seems to be all-important for the mechanism of crystal growth. Its existence has already been postulated by Marc (1908*b*, 1910, 1911*a*) and has been demonstrated by Volmer (1932) in the case of metal crystals growing from the melt or from the vapour.

There seem to be certain discrepancies in these results. One would be inclined to assume that the exchange of molecules between the solution and the adsorbed layer would be governed by laws of probability: the probability that a certain number of molecules goes over in a certain direction per unit time should be proportional to the number of collisions, i.e. to the concentration of the molecules, all other factors being equal. But here we find that *more* molecules go into the adsorbed layer at the centre of a face, where the concentration of the solution is *smaller* than at the corners, and where the concentration in the adsorbed layer must be greater than at the corners to produce the requisite lateral flow. If our interpretation of the concentration diagrams is correct the only solution out of this difficulty is to assume that there exists a hitherto unknown effect which makes the molecules in the solution and in the adsorbed layer react differently on the corners and at the centres of the faces. Lacking further experimental material no speculations will be made here as to the nature of this effect.

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Summary

An optical method of measuring the concentration distribution round a growing crystal is described. The flow of molecules at every point of the crystal surface can be determined. The results obtained on sodium chlorate are:

1—The concentration is not constant all over the crystal surface, but is highest at the corners.

2—One has to assume the existence of an adsorbed layer on the crystal surface, and that a lateral flow of molecules takes place in this layer.

3—There is a hitherto unexplained effect influencing the transition of molecules from the solution into the different parts of the adsorbed layer. The influx is smaller at the corners than at the centres of the faces, although the concentration of the solution is higher on the corners.

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